

a Range-guided retrieval and fusion

Query (Pb, ID 10)

Judge-context examples

Ranges (mg/g)

JEV judge

Output

Pyrolysis 500 °C
Solution pH 5.0
Initial C 40 mg/L
BET 208.4 m²/g

Q_e withheld

	T	pH	C ₀	BET	tBCpH	Q _e
Query						?
Ref 6						7.00
Ref 2						5.99
Ref 14						10.00
Ref 18						10.98
Ref 22						11.53
Ref 30						12.02
Ref 286						9.31
Ref 285						7.63

Query + 8 same-metal refs
Descriptors + measured capacity

B1: $0 \leq Q_e < 2$
B2: $2 \leq Q_e < 8$
B3: $8 \leq Q_e < 16$
B4: $16 \leq Q_e < 30$
B5: $30 \leq Q_e < 60$
B6: $Q_e \geq 60$

Fixed θ

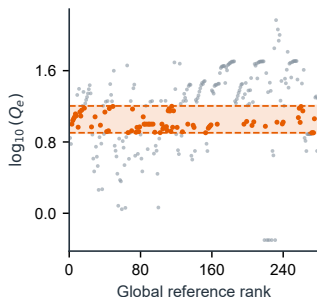
$p = [0, 0.07, 0.93, 0, 0, 0]$
 $c = 0.91$
 $b^* = 3$

$8 \leq Q_e < 16$

Selected range
(not final capacity)

Pb numerical reference pool

279 same-metal records

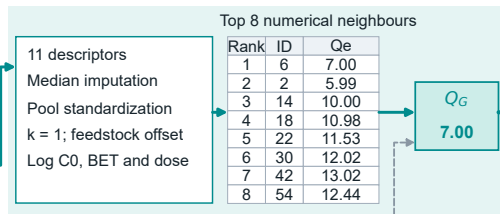


Grey: all other candidates

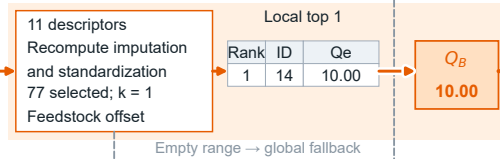
Orange: 77 in selected range

$$d_j = \sum_k [(\tilde{x}_k - \tilde{x}_{jk}) / \sigma_k]^2 - 1.5 I_{\text{same feedstock}}$$

Global retrieval: full library



Range-restricted retrieval



$$\log_{10} Q_F = 0.4 \log_{10} Q_B + 0.6 \log_{10} Q_G$$

$$Q_F = Q_B^{0.4} Q_G^{0.6}$$

Fusion

Fused estimate
8.07 mg/g